Higher National Diploma in Civil Engineering

04th Year, First Semester Examination - 2022

CE 3222 – Foundation Engineering

Instructions for Candidates:
Answer five questions only
Clearly mention any assumptions that you made

No. of questions: 06
No. of pages: 09
Time: 03 hours

Q1).

a) What are the four phases of sub soil exploration program? Briefly explain each phase. (4 marks)

b) What is meant by boring? Give the four types of boring? (4 marks)

c) State five information that should be included in the soil exploration report? (3 marks)

d) Name four factors that contribute to the variation of the Standard Penetration Test (SPT) results? (4 marks)

e) The blow count from the SPT at a depth of 7m from the ground surface (in loose sand layer) was found to be 10. The length of the drilling rod which is used to connect the split spoon sampler is 12m and no linear effect on the sampler. The diameter of the bore hole was found to be 180mm. The standard penetration test was performed with a standard split spoon sampler and hand dropped donut hammer with 60% energy efficiency. Calculate the corrected SPT - N value (N_{70}) at the depth of 7m below the ground surface. Refer Table Q1 for the correction factors. Consider that no effect from the water table. Following equation with usual notations may be useful in the calculations (Given: Density of the soil layer $\gamma = 18kN/m^3$, $1Pa = 1N/m^2$) (5 marks)

$$C_N = \sqrt{\frac{95.76}{\sigma'_v}}$$

$$N_{70} = N\eta_1\eta_2\eta_3\eta_4 C_N$$

Q2).

a) Using the phase diagram of soil elements show that the dry unit weight of the soil sample can be expressed as $\gamma_{dry} = \frac{(1+w)G_s\gamma_w}{(1+e)}$

G_s is the specific gravity of solid particles, w is the moisture content, γ is the unit weight of soil, e is void ratio and γ_w is the unit weight of water. (4 marks)

b) A soil sample having a total volume of $0.030m^3$ and total mass of 55.0 kg has been removed from the ground. If the water content and specific gravity of the soil are 25.0% and 2.68 respectively calculate:

i. The dry density of the sample
ii. The void ratio
iii. The degree of saturation
iv. The porosity

(8 marks)

c) State the objectives of following laboratory tests. (5 marks)

I. Grain size distribution test
II. Proctor compaction test
III. Atterberg limit test
IV. Cone penetration test
V. Vane shear test

d) State a test that can be conducted to estimate the pile capacity? Draw the schematic curve of Load Vs. Settlement for the above test for two consecutive cycles. (3 marks)

Q3).

a) Active earth pressure and passive earth pressure are two types of pressure components which should be considered when designing earth retaining structures. Explain the related terms with the aid of neat sketches where ever required. (4 marks)

b) Name four factors that should be considered when designing earth retaining structures? (4 marks)

c) A mass concrete retaining wall is 6m high. The retained soil material is in active state. Draw the effective horizontal stress and pore water pressure distributions on the retaining wall. Determine the Rankine active force and the location of the resultant. Check the wall

against overturning. All the required details are given in Figure Q3. Assume the wall surface is smooth. Assume the density of the concrete as 24 kN/m^3. No surcharge load on the soil surface. For sand, cohesion can be neglected.

(Hint: $\sigma_H = \sigma_v K_A - 2c\sqrt{K_A}$ and , $K_A = \frac{1-Sin\,\varphi}{1+Sin\,\varphi}$) (12 marks)

Q4).

a) What are the two methods to categorize piles? Give the types of piles under each category. (4 marks)

b) Describe about negative skin friction effect on a pile and explain three conditions where this effect can happen? (5 marks)

A pile has been driven for 10 m depth through two layers of soil having normally consolidated clay and dense sand. Refer to Figure Q4.1, Figure Q4.2 and Figure Q4.3. All the soil parameters are as given in the Figure Q4.1. Estimate,

i. Skin friction
ii. End bearing capacity
iii. Pile capacity (11 marks)

Use the following equations where ever necessary.

Unit skin friction: $q_s = \alpha C_u ----\alpha\ method$

$q_{av} = \lambda(\bar{\sigma'}_0 + 2C_u) ----\lambda\ method$

$q_s = \beta\sigma'_0 ----\beta\ method$

Total skin friction $Q_s = \int_0^L q_s\, p\, dz$

$\beta = k \tan\varphi$

$k = 1 - \sin\varphi$

End bearing capacity $q_p = C'N_C^* + q'N_q^*$

End bearing force $Q_p = A_p q' N_q^* \leq A_p q_1$

$q_1 = 0.5 p_a N_q^* \tan\varphi'$ } For sand

End bearing force $Q_p \approx N_C^* C_u A_p = 9C_u A_p$ ----For Saturated clay ($\varphi_u = 0$)

Q5).

a) What is known as a shallow foundation? Give three types of shallow foundations. (4 marks)

b) What are the shortcomings of Terzaghi's bearing capacity theory (4 marks)

c) A load 5° inclined to load is applied for a 2.0m wide strip footing as shown in Figure Q5. For a strip footing the length of the footing is considerably larger compared to the width. The embedment depth of the foundation is 2m, $\gamma = 19$ kN/m^3, C = 12kPa, $\varphi = 30°$. Consider the Factor of safety as 3. Refer to Table Q5 For the bearing capacity factors. Use the Meyerhof's equation to find,

   i. the ultimate bearing capacity
   ii. the gross allowable bearing capacity
   iii. the net ultimate bearing capacity
   iv. the net allowable bearing capacity (12 marks)

<u>Useful equations</u>

$$q_u = c'N_cF_{cs}F_{cd}F_{ci} + qN_qF_{qs}F_{qd}F_{qi} + \frac{1}{2}\gamma BN_\gamma F_{\gamma s}F_{\gamma d}F_{\gamma i}$$

Shape factors: $F_{cs} = 1 + \frac{B}{L}\frac{N_q}{N_c}$

$F_{qs} = 1 + \frac{B}{L}\tan\varphi'$

$F_{\gamma s} = 1 - 0.4\frac{B}{L}$

Depth factors: When $\frac{D_f}{B} \leq 1$,

$F_{cd} = 1 + 0.4\frac{D_f}{B}$

$F_{qd} = 1 + 2\tan\varphi'(1 - \sin\varphi')^2\frac{D_f}{B}$

$F_{\gamma d} = 1$

When $\frac{D_f}{B} > 1$,

$F_{cd} = 1 + 0.4tan^{-1}\left(\frac{D_f}{B}\right)$

$F_{qd} = 1 + 2\tan\varphi'(1 - \sin\varphi')^2\, tan^{-1}\left(\frac{D_f}{B}\right)$

$F_{\gamma d} = 1$

Inclination factors: $F_{ci} = F_{qi} = \left(1 - \frac{\beta}{90°}\right)^2$

$$F_{\gamma i} = \left(1 - \frac{\beta}{\varphi'}\right)^2$$

Q6).

a) Elaborate what are known as raft foundations with using sketches. (4 marks)

b) State four benefits of using raft foundations. (4 marks)

c) Briefly explain about Gabion walls. Your answer should include an introduction, benefits and applications of Gabion walls. (5 marks)

d) What is termed as internally stabilized earth supporting systems? Give three examples. (4 marks)

e) Determine the net ultimate bearing capacity of a mat foundation having 12m× 8m which has to be constructed on a saturated clay. For clay, $C_u = 95$ kN/m^2 and $\varphi = 0°$. D_f for the mat is 3m. Determine the net ultimate bearing capacity. (3 marks)

$$q_{(net)} = 5.14C_u\left(1 + \frac{0.195B}{L}\right)\left(1 + \frac{0.4D_f}{B}\right)$$

Table Q1: Correction factors

	Hammer for η_1				**Remarks**
	Donut		Safety		
Country	R-P	Trip	R-P	Trip/Auto	R-P = rope- pulley or cathead
United States					$\eta_1 = {E_r}/{E_{rb}}$
North America	45	-	70 - 80	70 - 100	For U.S trip/auto w/E_r = 80
Japan	67	78	-	-	$\eta_1 = \frac{80}{70} = 1.14$
United Kingdom	-	-	50	60	
China	50	60	-	-	
	Rod Length correction η_2				
Length	>10m		$\eta_2 = 1.00$		N is too high for L < 10m
	6-10m		$\eta_2 = 0.95$		
	4-6m		$\eta_2 = 0.85$		
	0-4m		$\eta_2 = 0.75$		
	Sampler correction η_3				
Without linear			$\eta_3 = 1.00$		Base value N is too high with linear
With linear : dense sand , clay			$\eta_3 = 0.80$		
With linear: loose sand			$\eta_3 = 0.90$		
	Borehole diameter correction η_4				
Hole diameter	60-100mm		$\eta_4 = 1.00$		Base value: N is too small when have an oversize hole
	150mm		$\eta_4 = 1.05$		
	200mm		$\eta_4 = 1.15$		

Table Q5: Bearing capacity factors for shallow foundations (Meyerhof equation)

ϕ	N_c	N_q	N_γ	N_q/N_c	tan ϕ	ϕ	N_c	N_q	N_γ	N_q/N_c	tan ϕ
0	5.14	1.00	0.00	0.20	0.00	26	22.25	11.85	12.54	0.53	0.49
1	5.38	1.09	0.07	0.20	0.02	27	23.94	13.20	14.47	0.55	0.51
2	5.63	1.20	0.15	0.21	0.03	28	25.80	14.72	16.72	0.57	0.53
3	5.90	1.31	0.24	0.22	0.05	29	27.86	16.44	19.34	0.59	0.55
4	6.19	1.43	0.34	0.23	0.07	30	30.14	18.40	22.40	0.61	0.58
5	6.49	1.57	0.45	0.24	0.09	31	32.67	20.63	25.99	0.63	0.60
6	6.81	1.72	0.57	0.25	0.11	32	35.49	23.18	30.22	0.65	0.62
7	7.16	1.88	0.71	0.26	0.12	33	38.64	26.09	35.19	0.68	0.65
8	7.53	2.06	0.86	0.27	0.14	34	42.16	29.44	41.06	0.70	0.67
9	7.92	2.25	1.03	0.28	0.16	35	46.12	33.30	48.03	0.72	0.70
10	8.35	2.47	1.22	0.30	0.18	36	50.59	37.75	56.31	0.75	0.73
11	8.80	2.71	1.44	0.31	0.19	37	55.63	42.92	66.19	0.77	0.75
12	9.28	2.97	1.69	0.32	0.21	38	61.35	48.93	78.03	0.80	0.78
13	9.81	3.26	1.97	0.33	0.23	39	67.87	55.96	92.25	0.82	0.81
14	10.37	3.59	2.29	0.35	0.25	40	75.31	64.20	109.41	0.85	0.84
15	10.98	3.94	2.65	0.36	0.27	41	83.86	73.90	130.22	0.88	0.87
16	11.63	4.34	3.06	0.37	0.29	42	93.71	85.38	155.55	0.91	0.90
17	12.34	4.77	3.53	0.39	0.31	43	105.11	99.02	186.54	0.94	0.93
18	13.10	5.26	4.07	0.40	0.32	44	118.37	115.31	224.64	0.97	0.97
19	13.93	5.80	4.68	0.42	0.34	45	133.88	134.88	271.76	1.01	1.00
20	14.83	6.40	5.39	0.43	0.36	46	152.10	158.51	330.35	1.04	1.04
21	15.82	7.07	6.20	0.45	0.38	47	173.64	187.21	403.67	1.08	1.07
22	16.88	7.82	7.13	0.46	0.40	48	199.26	222.31	496.01	1.12	1.11
23	18.05	8.66	8.20	0.48	0.42	49	229.93	265.51	613.16	1.15	1.15
24	19.32	9.60	9.44	0.50	0.45	50	266.89	319.07	762.89	1.20	1.19
25	20.72	10.66	10.88	0.51	0.47						

*After Vesic (1973)

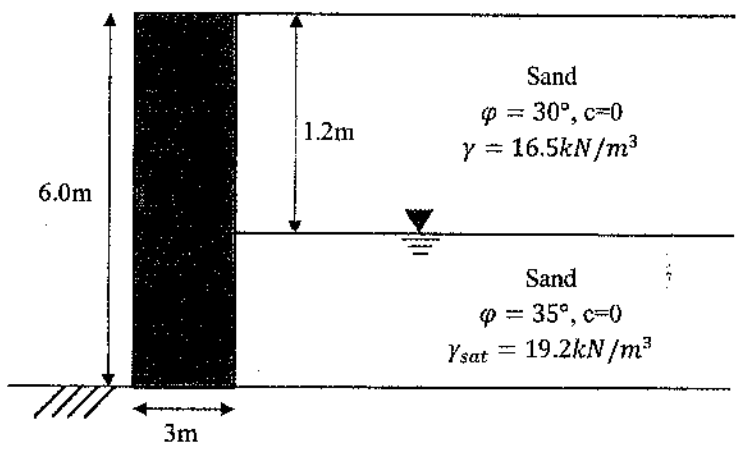

Figure Q3: Mass concrete wall for Q3

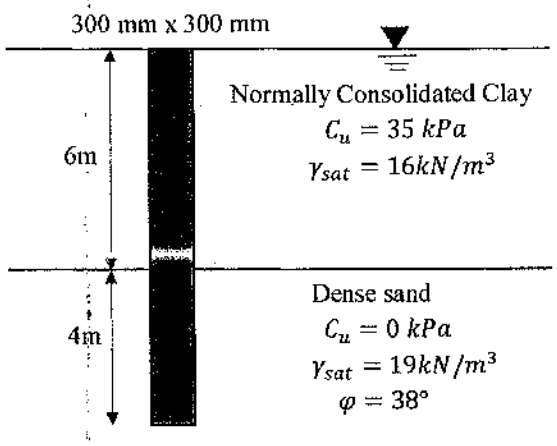

Figure Q4.1: Arrangement of the pile for Q4

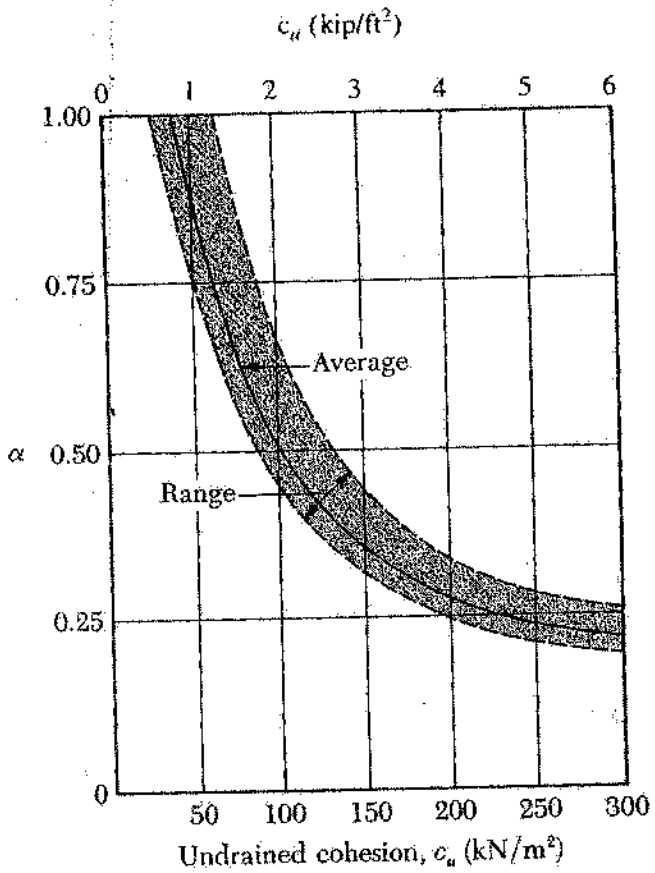

Figure Q4.2: Relationship between undrained cohesion and empirical adhesion factor

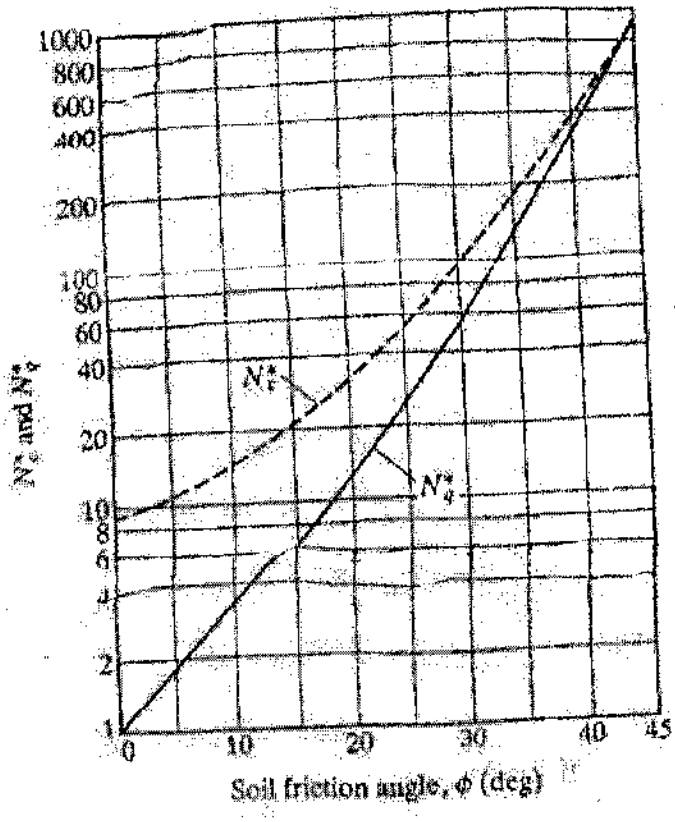

Figure Q4.3: Meyerhof's bearing capacity factors for piles

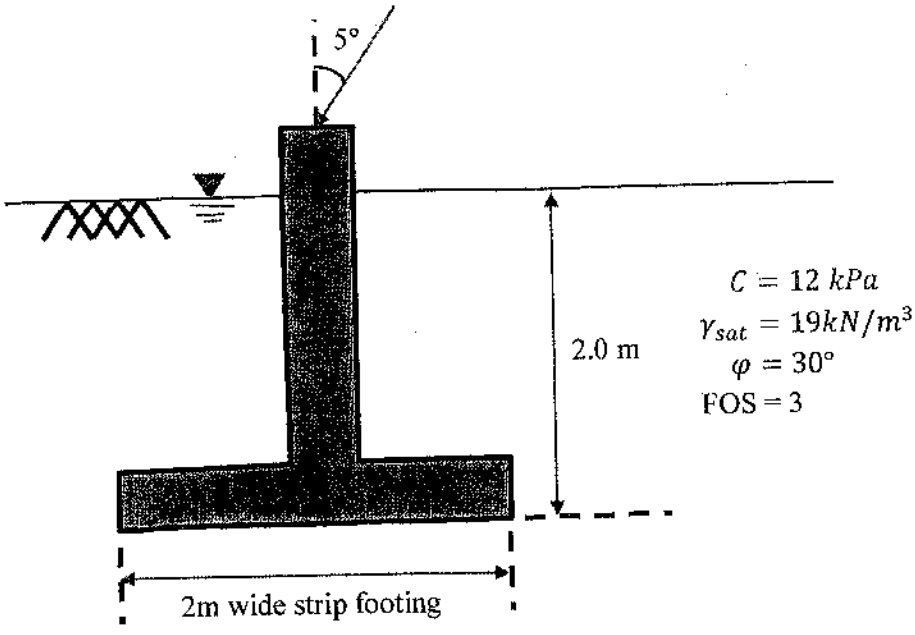

Figure Q5: Shallow foundation arrangement for Q5.